

# MakerWorld Quick Start Guide

## Braille Cylinder STL Generator — Parametric Model Maker edition

This guide walks you from "I want a braille label" to a downloadable STL pair using the single-file build in [makerworld/Braille Cylinder STL Generator MakerWorld v1.5.scad](#) on [MakerWorld's](#) Parametric Model Maker.

A PDF version of this guide is available: [MakerWorld Quick Start Guide.pdf](#).

## 1. What to Include

Braille takes far more space than print — you cannot include everything, so prioritize. The core question, per the Braille Authority of North America (BANA): "*Can someone identify me and contact me with just this information?*" If yes, you have enough.

- Each row holds **13 braille cells of text** by default. The generator offers **10 lines** (Line\_1–Line\_8 in the main tab, Line\_9–Line\_10 under **More Braille Lines (Advanced)**), though only the first grid\_rows of them are rendered — grid\_rows defaults to **4**, which is the usual business-card layout below. Raise it if you fill more lines; the generator warns you with TOO MANY LINES if you forget.
- BANA's typical four-line business card layout: **name, organization, phone number, e-mail address**. Job titles, mailing addresses, fax numbers, and websites are usually the first things to cut.
- If a name will not fit on one 13-cell line: drop capital indicators, drop a middle initial, use a first initial, or continue the name onto the next line.
- Organization names can be abbreviated (BANA's examples: lib for "library", amer for "American", nat for "National") or omitted entirely if the e-mail address already contains them.

Source: BANA *Business Cards Fact Sheet* (approved March 2024) — see Resources below.

## 2. Translate Your Text

MakerWorld's customizer cannot translate English for you. Paste **pre-translated Unicode braille**, one line per Line\_N field.

1. Open a braille translator such as <https://www.branah.com/braille-translator>.
2. Type your text in **lowercase** (each capital letter adds an indicator cell — braille readers understand omitting capitals is standard practice for space-limited labels).
3. Copy the **Unicode braille** output — the dot characters such as ⠠⠠⠠⠠⠠⠠ — not the ASCII/BRF letters.
4. Translate each line separately so you control where lines break.

Uncontracted (Grade 1) UEB is clearest for names and contact information.

## 3. Formatting

### Phone numbers

BANA guidance: braille a phone number as #123.456.7890 — omit parentheses and convert hyphens to periods. In Unified English Braille a period inside a number keeps *numeric mode* active, so only **one** number sign (⠠) is needed for the whole number. For example, 206.616.7678 translates to:

| ⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠ — exactly **13 cells**, which fits the default row.

**Why the hyphens have to go.** A period keeps numeric mode, but a **hyphen or parenthesis ends it**, so every hyphen-separated group needs a fresh number sign. 206-543-4779 translates to:

| ⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠ — **15 cells** with three ⠠, which will not fit.

That is correct UEB, not a translator bug: there is no setting anywhere that removes those signs. Retype

the number with periods and it drops to 13 cells, or delete the extra cells by hand before pasting.

Separately, some online translators repeat the number sign after each *period* too. That output is non-standard and wastes cells. So if your translation shows more than one  $\cdot$  in a phone number, check which cause you have: hyphens in the input (retype with periods) or a translator adding signs after periods (switch translators).

**Splitting a long number across two rows:** divide after a period and begin the next row with a fresh number sign:

Row 1:  $\cdot$   $\cdot$   $\cdot$   $\cdot$   $\cdot$   $\cdot$   $\cdot$   $\cdot$  (206.616.)

Row 2:  $\cdot$   $\cdot$   $\cdot$   $\cdot$   $\cdot$  (7678)

## E-mail and web addresses

Braille them per the Rules of UEB. When an address must be divided over two lines, divide after punctuation (the @ sign, a period, or a hyphen) and begin the second line in cell 1. A dot-5 line continuation indicator at the end of the first line is preferred; omit it only as a last resort. Tip: omit https:// or www. if the domain is clear without it.

## 4. Using the Customizer

1. Go to MakerWorld → **Create** → **Parametric Model Maker** and upload **only** Braille\_Cylinder\_STL\_Generator\_MakerWorld\_v1.5.scad.
2. In the parameter panel:
  - Paste your Unicode braille into Line\_1, Line\_2, Line\_3, Line\_4 (leave unused lines empty). Do **not** type plain English. Lines 5–8 are in the same tab and lines 9–10 under **More Braille Lines (Advanced)**; if you use any line past the fourth, raise grid\_rows to match or that text will not be rendered.
  - paper\_thickness\_preset: keep 0.4mm (default) for typical card stock, 0.3mm for thinner paper, or Custom to use your own slider values. Presets control dot, spacing and cylinder dimensions; they never change the grid\_columns / grid\_rows capacity sliders.
  - dot\_shape: Rounded (default) or Cone.
  - indicator\_mode: Visual (the Version 1 default) or Tactile (the Version 2 file's default since v2.10.0) — see section 5.
  - indicators (Indicator Letters): Visual mode only. On adds a square marker next to the triangle alignment indicator at each row's start. The triangle is always present — it is critical to the mechanical device the cylinder mounts into. On widens the grid by 2 marker cells, Off by 1 (triangle only) — your 13-cell text capacity is unchanged either way.
3. Generate and download the STL. By default one render builds **both cylinders** side by side — Cylinder A, the **Embossing Plate**, on the left and Cylinder B, the **Counter Plate**, on the right — so one download holds the pair. The two plates form a matching pair: paper goes between them, and pressing the embossing plate's dots into the counter plate's recesses forms readable braille.
4. To download the plates separately instead, set render\_both\_plates to Off, generate with plate\_type = Embossing Plate, then switch plate\_type to **Counter Plate** — change nothing else — and generate again.

## 5. Row Indicator Style: Visual or Tactile

indicator\_mode decides how each row is marked for alignment. The cylinder's size and cutout are the same either way — only the surface features change, so **generate both plates in the same mode**.

**Visual (default)** puts marker cells at the start of every row: a recessed triangle, plus a square when indicators is On. This is the long-standing layout and what the mechanical device expects.

**Tactile** removes those marker cells and puts one indicator per row in the seam gap instead — the empty stretch between the last and first cell as the text wraps around the cylinder. On the embossing plate it is a **raised arrow**; on the counter plate, a **matching recess** the arrow settles into as the pair rolls. On the **0.3mm** paper-thickness preset the cylinder carries exactly **three evenly spaced arrows** instead of one per row, so the two presets can be told apart by touch and a 0.3mm cylinder will not nest with a 0.4mm one.

Choose Tactile when a blind user needs to align the two cylinders unaided:

- The arrow feels like a single continuous wedge, nothing like a braille dot, and sits well clear of the nearest text.
- Its point faces the **top** of the cylinder, so you can tell which end is up by touch — on either plate.
- Raised on one plate, recessed on the other, so you can tell the embosser from the counter without sighted help.
- It is deliberately shallower than the braille dots (0.5 mm vs 1.0 mm), so the dots still do the embossing and the indicator leaves at most a faint mark.
- With no marker cells, up to **14 text cells** fit the default cylinder. The indicators toggle is ignored.

Five sliders tune the fit if you need them: `tactile_indicator_width`, `tactile_indicator_length`, `tactile_indicator_raise`, `tactile_recess_clearance`, and `tactile_recess_extra_depth`. The defaults are validated — raise the two recess values if the plates bind.

## 6. Cylinder Sizing

- **Measure the diameter** of the container you are labeling: wrap a string around it, measure the string, and divide by  $\pi$  (3.14159). Enter it as `cylinder_diameter_mm`, with `paper_thickness_preset` set to Custom — the 0.4mm and 0.3mm presets fix the diameter at 30.8 mm and ignore the slider.
- **Height:** measure the flat area where the braille will sit and leave a margin at the top and bottom.
- **Capacity math:** cells are spaced 6.5 mm apart, so the default 13 text cells + 2 indicator cells span about 91 mm. The default 30.8 mm-diameter cylinder (~96.8 mm circumference) fits that with room left for the seam gap. A larger diameter adds margin around the seam rather than more cells — raise `grid_columns` yourself if you want longer rows on a bigger cylinder. With Indicator Letters Off (triangle cell only) — or in Tactile mode, which has no marker cells — up to 14 text cells fit the default cylinder.
- Print cylinders standing upright for the best dot quality, and use `seam_offset_degrees` to rotate the seam away from your text.
- Leave your slicer's seam mode on **Aligned**: the shallow groove beside the row markers (the slicer seam channel, On by default) catches each layer's seam so it stays off the dots. A profile set to *Back* / *Rear* ignores the groove, so switch it once.

## 7. Troubleshooting

### INVALID CHARACTERS

Red 3D text reading INVALID CHARACTERS means a `Line_N` field contains something other than Unicode braille (typed English, ASCII braille, or stray spaces from another alphabet). Re-translate and copy the Unicode braille output.

### TEXT TOO LONG: 16/13

A line exceeds the row capacity. The warning shows the counts — longest line vs. capacity — because MakerWorld's preview cannot show console output. Fixes, in order of preference:

1. **Shorten or split** the line across rows (see the phone-number splitting pattern above).

2. **Raise grid\_columns** (Expert Mode - Braille Spacing section) if your cylinder is large enough.
3. **Set text\_limit\_check to Off** (Text Input section) to bypass the check entirely: every pasted cell renders and no warning appears, but rows longer than the capacity may crowd the seam gap.

### TACTILE GAP TOO SMALL: 5.8mm

Tactile mode only. The seam gap has shrunk below what the indicator plus a clear zone either side needs, so the arrow would crowd your braille text. The warning shows the measured gap and appears on both plates. At the default cylinder size the gap is 18.8 mm at 13 text cells and 12.3 mm at 14; 15 cells leaves only 5.8 mm and trips the warning. Fixes, in order of preference:

1. **Lower grid\_columns** back to 14 or fewer.
2. **Raise cylinder\_diameter\_mm** if you are labelling a larger container.
3. **Narrow tactile\_indicator\_width** (Row Indicator Style section) — but a narrower arrow is harder to find by touch, so treat this as a last resort.

### The plates bind or the indicator crushes the paper

- Binding: raise `tactile_recess_clearance` (outline margin) or `tactile_recess_extra_depth` (depth) so the arrow has more room.
- Heavy marking: lower `tactile_indicator_raise`. Keep it below the braille dot height so the dots, not the indicator, do the embossing.
- Very deep recesses thin the wall between the recess and the cylinder's internal cutout — at defaults that wall is about 1.22 mm, just above the 1.2 mm printable minimum, and `tactile_recess_extra_depth` eats into it directly.

## 8. Examples

Business-card examples take the content decisions from BANA's Fact Sheet (BANA's published braille is Grade 2; translate your own text as Grade 1).

| Scenario                               | Lines to translate and paste                                   | Notes                                                                                                 |
|----------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Name + e-mail (BANA Example 1)         | harry potter / harry@hogwarts.edu                              | Organization omitted — it already appears in the e-mail address.                                      |
| Long name (BANA Example 4)             | liesel a. / schimmelfennig / l.schimmelfennig@usace.army       | Name continues onto the second line; phone omitted.                                                   |
| Nickname + two phones (BANA Example 7) | fran rikard / albuquerque ac / c 505.312.4224 / f 505.312.4225 | Nickname saves cells; c/f prefixes tag cell and fax.                                                  |
| Spice jar (~55 mm diameter)            | cinnamon                                                       | One row; set <code>paper_thickness_preset</code> = Custom and <code>cylinder_diameter_mm</code> = 55. |
| Medicine bottle (~40 mm)               | amoxicillin / 500mg                                            | "amoxicillin" is 11 cells — within the 13-cell row.                                                   |
| Water bottle tag (~75 mm)              | j. smith / 555.867.5309                                        | Initials save space; the phone number fits one row.                                                   |

## 9. Resources

- [BANA Position Statements and Fact Sheets](#)
- [BANA Business Cards Fact Sheet \(PDF, approved March 2024\)](#)
- [BANA Size and Spacing of Braille Characters](#)
- [The Rules of Unified English Braille \(ICEB\)](#)
- [This project on GitHub](#)
- [Web app companion project](#) — translates English for you in the browser
- For complex cases (international phone numbers, multiple languages, credentials), consider working with a **UEB-certified transcriber**.